Fusion of Sparse Model Based on Randomly Erased Image for SAR Occluded Target Recognition

Zhiqiang He, Huaitie Xiao, Chao Gao, Zhuangzhuang Tian, and Si-Wei Chen, *Senior Member, IEEE*

***Abstract*— The recognition of partially occluded targets is a difficult problem in the field of synthetic aperture radar (SAR) target recognition. To eliminate the effect of occlusion, the intuitive idea is to determine the exact location and the size of the occluded area. However, this is very difficult, even impossible in practice. In order to avoid this difficulty and to improve the recognition performance for the partially occluded target, a fusion strategy of the sparse representation (SR) model based on randomly erased images is proposed to recognize the partially occluded target. The proposed method randomly erases some areas many times in both the test samples and the training samples. The erased training samples in each erasure are used to sparsely represent the corresponding erased test sample. Finally, all the SR results are fused to recognize the test sample. The proposed method utilizes random erasure to eliminate the possible occluded region. In addition, this method uses the fusion strategy to overcome under-erasing of the occluded region and erroneous erasure of the unoccluded region. The key parameter of the proposed method is the erasure ratio only. Although the erasure is random, the recognition performance of the method is relatively stable. Therefore, the method can eliminate the influence of occlusion without determining the details of occlusion. The experimental results show that the proposed method is significantly better than the state-of-the-art methods in the case of occlusion. Additionally, the recognition performance of the proposed method is similar to some comparison methods in the case of no occlusion.**

***Index Terms*— Automatic target recognition (ATR), fusion strategy, occluded target, random erasure, sparse representation (SR), synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) cannot be affected by weather and time. It plays an important role in military and civilian applications such as resource exploration, ocean observation, and target monitoring [1]–[3]. As an important aspect in the military application, SAR target detection and recognition have received extensive attention and research. Especially, the publication of the Moving and Stationary Target Acquisition and Recognition (MSTAR) [4] database promotes the development of SAR target recognition. Traditional SAR target recognition includes two aspects: feature extraction and recognition model.

The characteristics of the target include geometric feature [5], [6], transform domain feature, and electromagnetic scattering feature, and so on. The geometric features refer to the outline, corner, and shadow of the target. These features have clear physical meaning and play an important role in optical target recognition. However, there are significant differences between the SAR image and the optical image. It is often difficult to extract geometric features in SAR image. The transform domain features include principal component analysis (PCA) [7], [8], linear discriminant analysis (LDA), nonnegative matrix factorization (NMF) [9], wavelet transform (WT) [10], [11], monogenic signal [12], [13], and so on. These features can reflect different aspects of the target and play an important role in SAR target recognition. The electromagnetic scattering features include various electromagnetic scattering centers, such as the ideal point scattering center [14], [15], the attribute scattering center [16], [17], and so on. The extraction of the electromagnetic scattering center usually involves a complex mathematical model. In addition, the electromagnetic scattering feature is commonly used for template-based recognition.

The traditional recognition models include template-based recognition and model-based recognition [18]. Template-based recognition recognizes the target by comparing the similarity between the test sample and the templates. This method requires a complete template library, which is not only a huge workload, but also impossible in practical application. Model-based recognition establishes the mathematical model or the 3-D model of the target. The target is recognized according to the difference between the test sample model and the training sample model. With the development of computer technology and pattern recognition technology, machine learning method becomes the popular method in the field of target recognition due to its excellent performance. Typical machine learning methods include K-nearest neighbors (KNNs) [19], support vector machine (SVM) [20], [21], sparse representation (SR) [22]–[27], deep learning (DL) [28]–[31], and so on. In the standard operating condition (SOC), the DL has the best recognition performance. However, the DL requires a

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large amount of data to train the complex network. Different networks need to be trained for different application scenarios. In addition, there is no clear mathematical model to guide the design of the network. Personnel experience is important in the design process. The recognition performance of the KNNs, SVM, and SR under SOC is slightly lower than that of the DL. However, these methods require less data. Moreover, they have clear mathematical models. Therefore, it is impossible to arbitrarily judge the quality of these recognition models. It is necessary to select the most suitable model according to the application scenario.

By combining the feature and the recognition model, we can obtain SAR target recognition methods for various application scenarios. However, these methods usually take the complete target as the processing object. In practical application, the target is likely to be occluded by other objects. The target is occluded when some objects are in front of the target [14]–[16]. Some typical occlusion conditions include the target behind the bushes, the target behind the mound, and so on. In the occlusion condition, part of the target is covered by the occlusion object and only the rest of the target is visible to the radar. In the occluded target, the echo signal includes the echo signal of the target and the echo signal of the occlusion object. These two echo signals lead to some complex response, which causes the image of the occluded target to be different from the target image and the occlusion object image. Without considering the influence of the occlusion, the recognition performance can deteriorate seriously under the occlusion condition.

There are few articles on the occluded target recognition. Jones and Bhanu [14], [15] used the relative position of the scattering center to recognize the occluded target. Ding *et al.* [16] combined the attribute scattering center and the template matching to identify the occluded target. These methods must calculate the scattering centers of the target. Generally, the calculation model of the scattering center is very complex. Moreover, some parameters need to be set manually in the calculation process. Ding and Wen [32] combined the unoccluded target and the occluded target to improve the recognition performance. Strictly speaking, the occluded target only plays a supporting role in the method. This method is not specifically for the recognition of the occluded target. Wright *et al.* [22] proposed an improved the SR (ISR) model, which eliminates the effect of occlusion by introducing an error term into the SR model. He *et al.* [33] proposed the weighting method based on the subimage sparse model for SAR occluded target recognition. The method determines the possible occluded areas based on the sparse reconstruction errors of the subimages. In addition to the lack of articles, there is currently no publicly available database on the SAR image of the occluded target. The ideal synthesis method [14]–[16], [33] is used to obtain the SAR image of the occluded target in the occluded target recognition. In the ideal synthesis method, various SAR images of the occluded target are simulated by randomly using the background image to replace the target image. Although the image obtained by this method is different from the real image, it is also a reasonable simulation method [14], [15].

The difficulty of the occluded target recognition is to eliminate the influence of occlusion. The key is to determine the location and the size of the occluded area. In the absence of prior knowledge, it is often difficult or even impossible to accurately locate the position and the size of the occlusion. In order to improve the recognition ability under occlusion, we propose a fusion strategy of the SR model based on randomly erased images for SAR occluded target recognition. The proposed method utilizes random erasure to avoid determining the occlusion information. The random erasure means that the pixels of some areas in both the test samples and the training samples are zeroed to make them ineffective in recognition. The position of the erased area in the test sample is the same as that in the training samples. The effect of the occlusion can be perfectly eliminated under the condition that the erased area covers the occluded area. However, this ideal situation is usually not satisfied. In fact, the following situations often occur: the erased area can only cover part of the occluded area; even the erased area does not cover the occluded area at all. These conditions can cause some occluded areas or all occluded areas not to be zeroed. Moreover, some unoccluded areas are incorrectly zeroed. In order to eliminate the influence of these conditions on recognition, the proposed method adopts the strategy of multiple random erasures combined with decision-level fusion. The strategy performs multiple independent random erasures on the test samples and the training samples. Then, the sparse reconstruction errors of the erased test samples on the corresponding erased training samples are calculated. Finally, the proposed method fuses these sparse reconstruction errors. The recognition is achieved by the fused sparse reconstruction error. The key parameter of the proposed method is the size of the erased area (erasure ratio) in single erasure. Although the proposed method cannot guarantee that single erasure can completely cover the occluded area, the probability that all the erased areas completely cover the occluded area can significantly improve by setting the appropriate erasure ratio and performing multiple random erasures. The useful information which is erased by mistake in single erasure can be compensated in other erasures. The fusion strategy can reduce the information loss caused by the false erasure in single erasure. In summary, multiple random erasures combined with decision-level fusion can improve the recognition performance under occlusion. To this end, the main contributions are as follows.

1) The target recognition model based on the SR uses the sparse reconstruction error to determine the class of the test sample. The reconstruction error of the test sample is the sum of the reconstruction errors of all elements of the test sample. Therefore, the influence of the occlusion can be eliminated by setting the sparse reconstruction error corresponding to the occluded part of the test sample to zero. Based on this principle, the proposed method uses random erasure to eliminate the occlusion. The reconstruction error of the pixels in the zeroed region is zero, so these pixels do not play a role in recognizing the target. Although the possibility that the erased area and the occluded area completely coincide

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is small, the random erasure can completely eliminate the occlusion information under the condition that the erased area and the occluded area completely coincide.

2) Since the erased area is randomly selected, there is little possibility that the erased area and the occluded area completely coincide. The common situations are under-erasure of the occluded area and erroneous erasure of the unoccluded area. In order to overcome the influence of these situations on recognition performance, the proposed method proposes the strategy of random erasure combined with decision-level fusion. The fusion strategy can increase the probability that all erased areas of these random erasures cover the occluded area. The target information that is erroneously erased in one random erasure can be compensated in other random erasures. Therefore, the combination of random erasure and decision-level fusion can eliminate the occlusion while retaining the target information.
3) The effects of the erasure types and the fusion strategies on recognition performance are verified through some experiments. The optimal erasure type and the optimal fusion strategy are obtained. Some application scenarios are designed in this article. The statistical characteristics of the proposed method in these scenarios are analyzed. The recognition performance of the proposed method and some traditional methods is compared. The experimental results show that the performance of the proposed method to recognize the occluded target is better than that of the traditional methods.

The rest of this article is organized as follows. Section II describes the traditional SR model. Section III proposes the fusion strategy of SR model based on randomly erased SAR images. Section IV describes the experimental results on MSTAR database. Finally, the conclusion is given in Section V.

II. SR Model

As the combination of compressed sensing and machine learning, the SR is widely used in the fields of target imaging, detection, and recognition. The SR model uses the linear combination of the dictionary atoms to represent the optimal approximation of the test sample. In the SR recognition model, the test sample can be sparsely represented by the dictionary atoms of the same class. The sparse reconstruction error is small. But the test sample cannot be sparsely represented by the dictionary atoms of different class. The sparse reconstruction error is large. The test sample is denoted as: $\mathbf{y} = [y_1, y_2, y_3, \ldots, y_N]^T \in \mathbb{R}^N$, where $y_i, i \in [1, N]$ is the ith element of the test sample, and N is the dimension of the test sample. The dictionary is denoted as $\mathbf{D} = [\mathbf{D}_1, \mathbf{D}_2, \ldots, \mathbf{D}_C] \in \mathbb{R}^{N\times M}$, where $\mathbf{D}_i = [\mathbf{d}_{i,1}, \mathbf{d}_{i,2}, \ldots, \mathbf{d}_{i,n_i}] \in \mathbb{R}^{N\times n_i}$ is the ith subdictionary, $\mathbf{d}_{i,j}$, $j \in [1, n_i]$ is the jth atom of the ith subdictionary, n_i is the atom number of the ith subdictionary, $M = \sum_{i=1}^{C} n_i$ is the total atom number, C is the class number. Usually the dictionary is overcomplete, i.e., $M \gg N$. SR is denoted as

$$\mathbf{y} = \mathbf{D}\mathbf{x} \tag{1}$$

where $\mathbf{x} = [x_1, x_2, x_3, \ldots, x_M]^T \in \mathbb{R}^{M\times 1}$ is the projection vector (sparse coefficient vector) of the test sample $\mathbf{y}$. Since the dictionary $\mathbf{D}$ is overcomplete, the number of $\mathbf{x}$'s that meet the requirement is infinite. In fact, it is more meaningful to calculate $\mathbf{x}$ with the smallest number of nonzero elements. In this case, the objective function of the SR is denoted as

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}} \|\mathbf{y} - \mathbf{D}\mathbf{x}\|_2, \quad \text{s.t. } \|\mathbf{x}\|_0 \le K \tag{2}$$

where K is the sparsity. The above formula is an NP-hard problem. It needs to be solved by the greedy algorithm or the convex relaxation algorithm (in this case, the norm $\|\mathbf{x}\|_0$ needs to be replaced by the norm $\|\mathbf{x}\|_1$). After obtained the sparse coefficient vector $\hat{\mathbf{x}}$, the sparse reconstruction vector of the test sample $\mathbf{y}$ is $\hat{\mathbf{y}} = \mathbf{D}\hat{\mathbf{x}}$, and the sparse reconstruction error is $e = \|\mathbf{y} - \hat{\mathbf{y}}\|_2$.

Let $\delta_i : \mathbb{R}^N \to \mathbb{R}^N$ be the characteristic function that selects the coefficients associated with the ith class. The sparse reconstruction vector of the test sample $\mathbf{y}$ on the ith subdictionary $\mathbf{D}_i$ is denoted as $\hat{\mathbf{y}}_i = \mathbf{D}\delta_i(\hat{\mathbf{x}})$. The corresponding ith sparse reconstruction error is $e_i = \|\mathbf{y} - \hat{\mathbf{y}}_i\|_2$. The class of the test sample $\mathbf{y}$ is the class of the subdictionary corresponding to the smallest sparse reconstruction error

$$\text{class}(\mathbf{y}) = \min_i \|\mathbf{y} - \hat{\mathbf{y}}_i\|_2. \tag{3}$$

III. Fusion of SR Based on Randomly Erased SAR Image

It can be seen from the definition of the sparse reconstruction error that the sparse reconstruction error e of the test sample $\mathbf{y}$ can also be expressed as

$$e^2 = \sum_{i=1}^{N} e_{(i)}^2 \tag{4}$$

where $e_{(i)} = y_i - \sum_{j=1}^{M} d_{(i,j)}x_j$ is the sparse reconstruction error of the ith element of the vector, and $d_{(i,j)}$ is the element of the dictionary $\mathbf{D}$ at the coordinate (i, j). The above equation indicates that the sparse reconstruction error of the test sample $\mathbf{y}$ is the sum of the sparse reconstruction errors of all elements of the test sample. The SR model uses the sparse reconstruction error as the identification index. To make the subsample $\mathbf{y}^s = [y_{s1}, y_{s1+1}, \ldots, y_{s1+N^s}]^T \in \mathbb{R}^{N^s}$ ineffective in recognition, it is necessary to set its sparse reconstruction error to zero. According to the definition of the sparse reconstruction error, this requires that both the elements $y_i, i \in [s1, s1 + N^s]$ in the test sample and the elements $d_{(i,j)}, i \in [s1, s1 + N^s]$ in the dictionary are set to zero.

The condition that the test sample $\mathbf{y}$ is partially occluded by the occlusion object is equal to the condition that a certain subvector $\mathbf{y}^o$ in $\mathbf{y}$ is replaced with an occlusion vector $\mathbf{v}$, where $\mathbf{y}^o, \mathbf{v} \in \mathbb{R}^{N^o\times 1}$, and N^o is the size of the occluded region. Assuming that the subdictionary corresponding to the subvector $\mathbf{y}^o$ is denoted as $\mathbf{D}^o$, the relationships between the subvector $\mathbf{y}^o$ and the vector $\mathbf{y}$, the subdictionary $\mathbf{D}^o$ and the dictionary $\mathbf{D}$ are shown in Fig. 1. The corresponding sparse reconstruction errors $e^{(\mathbf{y}^o)}$, $e^{(\mathbf{v})}$ are denoted as

$$e^{(\mathbf{y}^o)} = \|\mathbf{y}^o - \mathbf{D}^o\hat{\mathbf{x}}\|_2 \tag{5}$$

$$e^{(\mathbf{v})} = \|\mathbf{v} - \mathbf{D}^o\hat{\mathbf{x}}\|_2. \tag{6}$$

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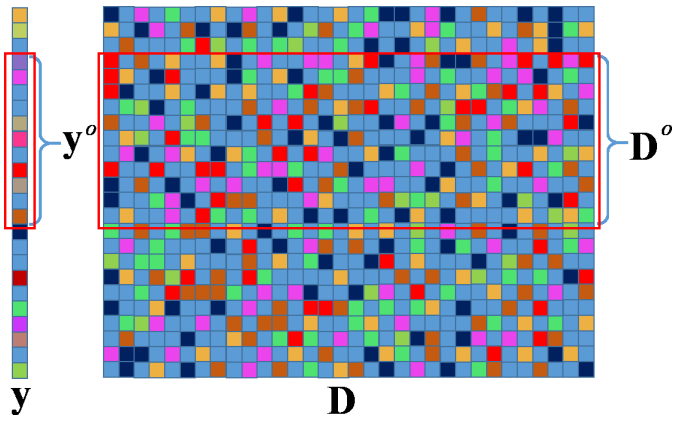


Fig. 1. Relationship between the vector and the subvector, the dictionary and the subdictionary.

In the absence of occlusion, the test sample $\mathbf{y}$ and some atoms of the dictionary $\mathbf{D}$ belong to the same class. The subvector $\mathbf{y}^o$ and the corresponding atoms of the subdictionary $\mathbf{D}^o$ also belong to the same class. At this time, the sparse reconstruction error $e^{(\mathbf{y}^o)}$ is small. In the occlusion case, the subvector $\mathbf{v}$ and the subdictionary $\mathbf{D}^o$ do not belong to the same class. At this time, the sparse reconstruction error $e^{(\mathbf{v})}$ usually is large. This means that replacing $\mathbf{y}^o$ with $\mathbf{v}$ can result in a change in the total sparse reconstruction error. Since the SR classifier uses the sparse reconstruction error as the identification index, the occlusion of the test sample can affect the performance of this recognition method.

To eliminate the influence of occlusion on recognition, it is necessary to remove the sparse reconstruction error $e^{(\mathbf{v})}$ corresponding to the occlusion vector $\mathbf{v}$ from the sparse reconstruction error e. This means that the sparse reconstruction error corresponding to the occlusion vector needs to be set to 0. This process needs to determine the position and the size of the occlusion vector $\mathbf{v}$. The idea is correct, but it is very difficult to implement. The difficulties mainly include the following aspects: First, the prior knowledge of the occlusion is difficult to obtain. Then the exact position of the occlusion object cannot be confirmed. Second, the shape of the occlusion object may be irregular. Then it is difficult to determine the size of the occlusion object. Finally, the SAR image of the occluded target is not simply replacing the target image with the occlusion object image. There is some complex interaction between the occlusion object and the target. The boundary between the occlusion object image and the target image may not be clear.

Since it is very difficult to determine the position and the size of the occlusion object, we use random erasure on the test sample and the dictionary to avoid this processing. Random erasure refers to zeroing some areas of the sample. The zeroed areas are randomly selected. The erasure areas of the test sample and the dictionary atoms are the same. The example of the random erasure is shown in Fig. 2. The erased test sample is $\mathbf{y}'$. The erased dictionary is $\mathbf{D}'$. The corresponding sparse coefficient vector is $\hat{\mathbf{x}}'$. At this time, the reconstruction error vector of the erased test sample $\mathbf{y}'$ on all subdictionaries is $\mathbf{e}' = [e'_1, e'_2, e'_3, \ldots, e'_C]^T$, where $e'_i = \|\mathbf{y}' - \mathbf{D}'\delta_i(\hat{\mathbf{x}}')\|_2, i \in [1, C]$. Ideally, the erased area is exactly the occluded area. Therefore, the class of the test sample $\mathbf{y}$ is the class of the subdictionary corresponding to the minimum sparse reconstruction error

$$\text{class}(\mathbf{y}) = \min_i \left\{e'_i, i \in [1, C]\right\}. \tag{7}$$

This ideal condition is usually impossible to meet in practice. When the erased area does not exactly match the occluded area, the following problems are caused. The first one is that the occluded area cannot be completely erased. The second one is that some unoccluded area is erased by mistake. In this section, we use multiple random erasures and decision-level fusion to overcome these problems. First, we perform L random erasures on the test sample $\mathbf{y}$ and the dictionary $\mathbf{D}$, respectively. Then, we use the erased test sample and the corresponding erased dictionary to calculate the reconstruction error vector set $\Phi = \{\mathbf{e}'^{(i)}\}, i \in [1, L]$. Finally, the decision-level fusion is performed on the reconstruction error vector set Φ to obtain the fusion result $\mathbf{e}_f = [e_f^1, e_f^2, e_f^3, \ldots, e_f^C]^T$, where e_f^i is the fusion result of ith sparse reconstruction error. We take the arithmetic average (AAv) fusion algorithm as an example. The fusion result at this time is denoted as $\mathbf{e}_f = (1/L)\sum_{i=1}^{L} \mathbf{e}'^{(i)}$. The class of the test sample $\mathbf{y}$ is the class corresponding to the minimum fusion result

$$\text{class}(\mathbf{y}) = \min_i \left\{e_f^i, i \in [1, C]\right\}. \tag{8}$$

The shortcomings of single random erasure have described above. Next, we discuss the reasons why multiple random erasures combining with decision-level fusion can overcome these shortcomings. A single erasure may eliminate some occluded area by an appropriate erasure ratio. The erasure ratio is the ratio between the size of the erased area and the size of the target image. Usually, the occluded area eliminated in each random erasure is different. It can combine the occluded areas eliminated in these random erasures by fusing the results of these random erasures. Therefore, multiple random erasures can effectively avoid the problem of incomplete erasure of occluded area. Since the erased area is completely randomly selected, the erasure operation inevitably causes some unoccluded areas to be erroneously erased. The reasons why random erasure combining with fusion strategy can overcome this problem are as follows.

First, the pixels of the image have different effects on recognition. Some pixels play a key role, while the others play a minor role. We can call these pixels which have little effect on recognition as redundant pixels. This is also the reason that reasonable dimensionality reduction does not affect the recognition performance [34]. The erroneously erased pixels may include some redundant pixels. Although these pixels are erroneously erased, they have less influence on the recognition performance.

Second, the occluded area and the unoccluded area have different effects on the recognition performance. The unoccluded area provides correct information on recognition. The error erasure of the unoccluded area only reduces the correct information. The occlusion not only eliminates the original

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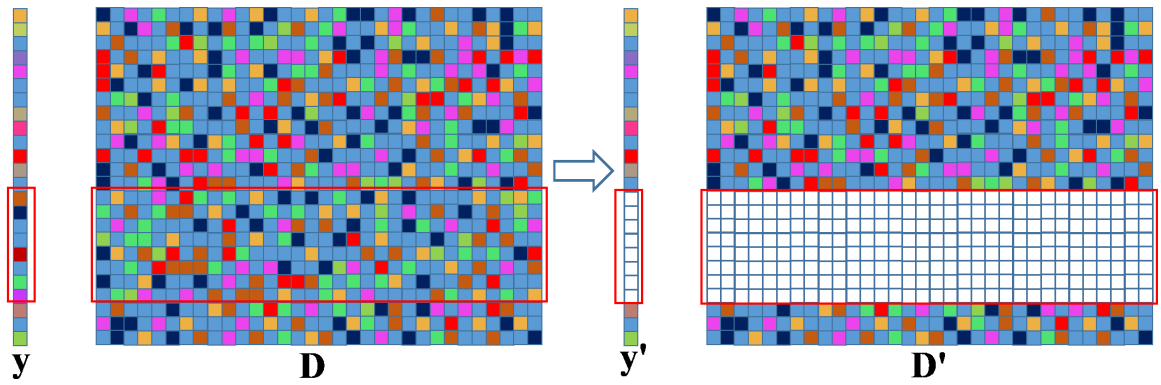


Fig. 2. Example of random erasure. $\mathbf{Y}$ and $\mathbf{D}$ are the original test sample and original dictionary, $\mathbf{y}'$ and $\mathbf{D}'$ are the erased test sample and erased dictionary. The area in the red box is the erased area. The position of the erased area in the test sample is the same as that in the training sample.

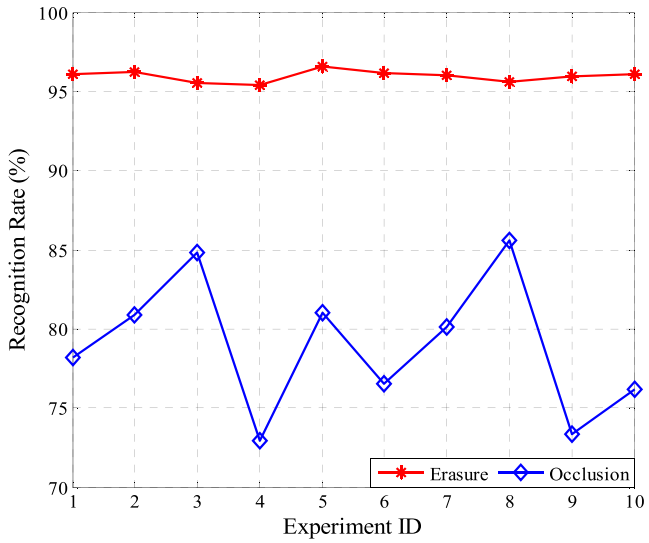


Fig. 3. Recognition rates of erased image and occluded image.

correct information in the occluded area, but also introduces some error information. Usually, the effect of the error information on recognition is more serious than that of the absence of the correct information. We prove this principle through some experiments. The test images in these experiments are divided into randomly erased images and occluded images. The erasure ratio is 0.3. The occluded area is exactly the same as the erased area. The parameters of these experiments are the same as those of Section IV-B. We conduct ten experiments independently. The experimental results are shown in Fig. 3. It can be seen from the experimental results that the recognition rate of the erased images is significantly better than that of the occluded image. Therefore, the benefit of erasing the occluded area is greater than the disadvantage of erroneously erasing the unoccluded area. In other words, the over-erasing of the unoccluded area is better than the under-erasing of the occluded area.

Finally, although single random erasure may cause some unoccluded area to be erased by mistake, the information that is erroneously eliminated in one erasure may be compensated in other erasures.

In summary, random erasure combining with fusion strategy can not only eliminate the influence of the occlusion, but also reduce the impact of false erasure on recognition performance. The flow of the fusion strategy of SR model based on randomly erased SAR images is shown in Algorithm 1.

Algorithm 1 FUSION ALGORITHM BASED ON RANDOM ERASURE

Input: the test image (test sample) $\mathbf{A}$, the train image (train sample) set $\{\mathbf{B}_i\}, i \in [1, M]$, erasure times L, erasure ratio set $\{r_i\}, i \in [1, L]$, sparsity K, current operation count $l = 1$.
Output: The class of the test image.
Algorithm Flow:
1. **While** $l \leq L$ **do**
2. A square area whose size is r_l times of the target image is randomly erased in both the test image $\mathbf{A}$ and the training image set $\{\mathbf{B}_i\}$. These erased images are vectorized into the test sample $\mathbf{y}'_l$ and the dictionary $\mathbf{D}'_l$.
3. Calculate the sparse coefficient vector $\hat{\mathbf{x}}'$ of the test sample $\mathbf{y}'_l$ on the dictionary $\mathbf{D}'_l$. According to the sparse coefficient vector, calculate the sparse reconstruction error vector $\mathbf{e}'^{(l)}$.
4. $l = l + 1$
5. **end**
6. The fused sparse reconstruction error vector $\mathbf{e}_f$ is obtained by fusing the sparse reconstruction error vector set Φ obtained by L times random erasures. Then, the class of the test image $\mathbf{A}$ is the class corresponding to the minimum element of the sparse reconstruction error vector $\mathbf{e}_f$.
7. Output the recognition result.

Next, we analyze the complexity of the proposed method. Assuming that the erasure number of the proposed method is L, the processing of the proposed method includes L random erasures, calculating L sparse coefficient vectors and sparse reconstruction error vectors, and fusing these reconstruction error vectors. Among these steps, calculating the sparse coefficient vector and reconstruction error vector are the most time-consuming step. Therefore, the complexity of the proposed method is mainly determined by calculating the sparse coefficient vector and the reconstruction error vector. We use the orthogonal matching pursuit (OMP) algorithm

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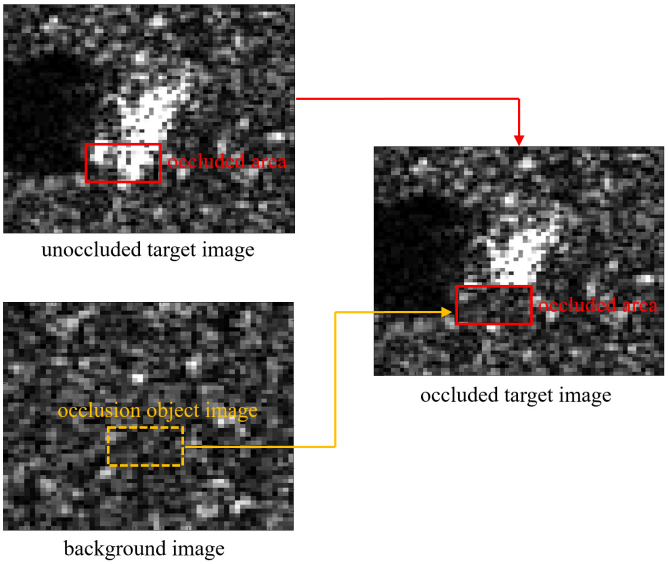


Fig. 4. Process of generating the occluded target image by the ideal synthesis method. The area in the red box is the occluded area. The area in the yellow dashed box is the occlusion object image, which is randomly selected in the background image.

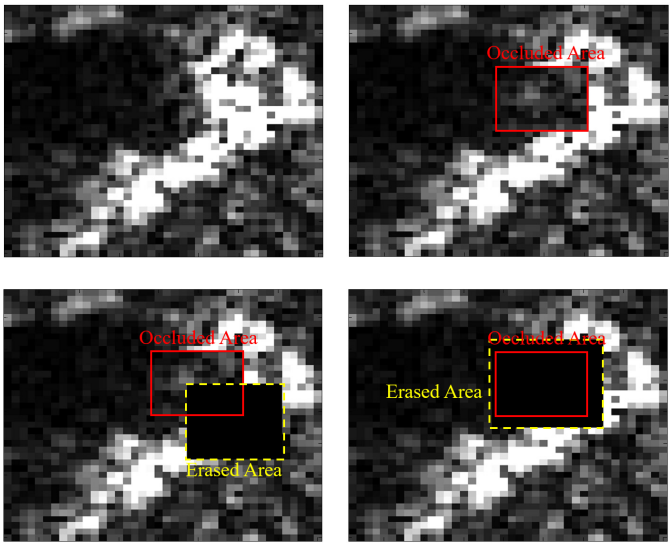


Fig. 5. Example of occlusion and random erasure. The first line from left to right shows the original image and the occluded target image. The second line from left to right shows the erased image under the condition that the erased area is not aligned with the occluded area, and the erased image under the condition that the erased area is aligned with the occluded area. The area in the red box is the occluded area. The area in the yellow dotted box is the erased area.

to calculate the sparse coefficient vector of the test sample in the proposed method. The complexity of the OMP is $O(KMN)$, where K is the sparsity, M is the atom number of the dictionary, and N is the dimension of the test sample. The proposed method needs to execute the OMP algorithm L times. Then, the complexity of the proposed method is $O(LKMN)$.

IV. Experiment and Discussion

A. Data Preparation

In this section, we use the MSTAR database to evaluate the recognition performance of the proposed method. The MSTAR database contains ten military targets. These targets are two tanks (T72, T62), four armored personnel carriers (BTR70, BTR60, BRDM2, and BMP2), a truck (ZIL131), a cannon (2S1), an air defense unit (ZSU234), and a bulldozer (D7). The optical images of these targets have significant differences. However, their SAR images are similar and difficult to distinguish by the naked eye. The resolution of the image is 0.3 m $\times$ 0.3 m. The depression angle of the radar includes 15°, 17°, and 30°. The aspect angle of the target is 0°–360°. The difference of aspect angles in adjacent images is 1°–3°. Each SAR image in the database contains one target. The size of the image is 128 $\times$ 128 pixels. For the convenience of processing, we cut the 64 $\times$ 64 pixels slice in the center of the image as the processing object. In the experiments, we randomly erase the image slices. The test samples and the training samples are obtained by down-sampling the erased image slices. The dimension of these samples is 1024.

In the experiments, we use the ideal synthesis method to obtain the occluded target image. We set the occlusion ratio from 0.0 to 0.5, where the occlusion ratio of 0.0 means no occlusion. The occlusion area is square. The process of generating the occluded target image by the ideal synthesis method is shown in Fig. 4.

The operating condition for target recognition includes the SOC and the extended operating condition (EOC). The SOC means that the class and the serial number between the training samples and the test samples are the same. The number of the training samples is sufficient. The test samples are not contaminated by noise. Correspondingly, the case where the SOC is not met is called the EOC. The experiments are divided into two types: the experiment in SOC and the experiments in EOC. The experiment in SOC takes ten classes target as the experimental objects. The experiments in EOC include five parts: incomplete training samples, random noise corruption of the test samples, low resolution, configuration variance, and depression angle variance.

In the experiments, the erasure types include many square areas with different erasure ratio and some combinations of these square areas. The erasure type and its ID are shown in Table I. An example of occlusion and random erasure is shown in Fig. 5. We use linear SVM (LSVM), kernel SVM (KSVM), traditional SR (SR), kernel SR (KSR), ISR [22], weighted SR [33], convolutional neural network (CNN) [35], decision-level fusion with LSVM and KSVM (FSVM), and decision-level fusion with SR and KSR (FSR) as the comparison algorithms. The parameters related to the experiments are as follows: the sparsity $K = 10$, the erasure times $L = 20$. Unless otherwise specified, the proposed method selects the AAv method as the fusion algorithm.

B. Recognition of Ten Classes Target in SOC

This experiment selects all ten classes of the targets in MSTAR database as the experimental objects. The serial numbers of the training samples and the test samples are the same. For BMP2 and T72 which have various serial numbers,

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TABLE I
ERASURE RATIO AND ERASURE NUMBER OF ERASURE TYPE

ID	Erasure type	Erasure number
I	0.1	20
II	0.2	20
III	0.3	20
IV	0.4	20
V	0.5	20
VI	0.1+0.2	10+10
VII	0.1+0.2+0.3	6+7+7
VIII	0.1+0.2+0.3+0.4	5+5+5+5
IX	0.1+0.2+0.3+0.4+0.5	4+4+4+4+4
X	0.2+0.3	10+10
XI	0.2+0.3+0.4	6+7+7
XII	0.2+0.3+0.4+0.5	5+5+5+5
XIII	0.3+0.4	10+10
XIV	0.3+0.4+0.5	6+7+7
XV	0.4+0.5	10+10

TABLE II
STATISTICAL INFORMATION OF TEN CLASSES TARGET

Class	Train(17°)	Test(15°)
BMP2 (sn-c9563)	233	195
BTR70	233	196
T72 (sn-132)	232	196
BTR60	256	195
2S1	299	274
BRDM2	298	274
D7	299	274
T62	299	273
ZIL131	299	274
ZSU23/4	299	274

the serial numbers sn-c9563 (BMP2) and sn-132 (T72) are used as the experimental objects. The images captured at depression angle 17° are the training samples. The images captured at depression angle 15° are the test samples. The statistical information of the experimental data is shown in Table II.

1) Recognition Results of Different Erasure Types: The recognition results of the proposed method with different erasure types are shown in Table III. When the occlusion ratio is 0.0, 0.1, and 0.2, the recognition rates of all erasure types are higher than 97.00%, 96.00%, and 94.00%, respectively. This indicates that all erasure types have good recognition performance under the condition of small occlusion ratio. When the occlusion ratio is greater than 0.3, the recognition performance of different erasure types is significantly different. The erasure types I, II, VI, and VII have poor recognition performance. However, the erasure types IV, V, XIV, and XV have better recognition performance. We can find out that the erasure types with poor recognition performance include small erasure ratio (0.1, 0.2). The erasure types with good recognition performance only include some large erasure ratio. This indicates that large erasure ratio favors good recognition performance. The reason is that the large erasure ratio can increase the probability that the erased area covers the occluded area. There are two reasons for the poor recognition rate of small erasure ratio. The one is that the probability of the erased area aligning with the occluded area is small. The other is that the erased area can not completely cover the occluded area, even if the erased area aligns with the occluded area. When the occlusion ratio is small, the large erasure ratio also maintains good recognition performance. The reason is that the fusion of multiple erasures can effectively compensate for the information that is erroneously erased in single erasure. From the above analysis, it is also known that over-erasing is superior to under-erasing in single erasure. In subsequent experiments, we choose the erasure type XV to verify the performance of the proposed method.

2) Recognition Results of Different Fusion Methods: The proposed method uses the decision-level fusion strategy to fuse all sparse reconstruction error vectors of the erased images. In addition to the AAv method, there are many decision-level fusion methods, such as random weighting average (RWAv) method, Dempster–Shafer (DS) method [36], ranking-based (RB) method [37], and so on. The fusion performance and applicable scenario of these fusion methods are different. In this experiment, we verify the performance of various fusion methods in occluded target recognition. The recognition results with various fusion methods are shown in Table IV.

When the occlusion ratio is less than 0.2, the recognition performance of various fusion methods is basically similar. When the occlusion ratio is greater than 0.2, the recognition performance of the AAv, RWAv, and DS methods is similar, while the recognition rate of the RB method is slightly lower than that of the other methods. However, the difference among the performance of these methods is very small. This indicates that the fusion method has less impact on recognition performance.

3) Statistical Property of the Proposed Algorithm: Because the erased area of the proposed method is random, the statistical property of the proposed method is an important index. This section verifies the statistical property of the proposed method through multiple experiments. We use the mean and the variance as the indicators. Assume that the recognition rate vector of multiple recognition experiments is $\mathbf{P} = \{p_1, p_2, p_3, \ldots, p_J\}$, where J is the number of the experiments, and $p_i, i \in [1, J]$ is the recognition rate of the ith experiment. Then, the mean and the variance are defined as

$$\text{mean} = \left(\sum_{j=1}^{J} p_j \right) / J \tag{9}$$

$$\text{var} = \frac{1}{J} \left(\sum_{j=1}^{J} (p_j - \text{mean})^2 \right). \tag{10}$$

The smaller the variance, the better the stability of the proposed method, and vice versa. We perform 100 experiments using the erasure type XV. The experimental results are shown in Table V and Fig. 6. We also list the maximum recognition rate and the minimum recognition rate in Table V.

Although the variance has an increasing trend as the occlusion ratio increases, the maximum value of the variance is 0.299. The absolute difference of the recognition rates between the maximum recognition rate and the minimum recognition rate is 2.65%. Even with the worst recognition performance, the recognition rate at the occlusion ratio of 0.5 is higher

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TABLE III
RECOGNITION RESULTS WITH DIFFERENT ERASURE TYPES

Occlusion Ratio	Erasure types														
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI	XII	XIII	XIV	XV
0.0	97.69	97.65	97.59	97.23	97.13	97.90	97.72	97.69	97.67	97.68	97.71	97.65	97.43	97.59	97.44
0.1	96.56	97.04	96.91	96.97	96.70	97.01	97.36	97.49	97.47	97.28	97.41	97.40	97.38	97.38	97.19
0.2	94.10	95.48	95.95	96.09	96.46	95.17	95.74	96.34	96.61	95.97	96.35	96.54	96.53	96.55	96.69
0.3	85.28	90.26	93.59	95.17	95.19	88.20	91.73	93.40	94.41	93.62	94.00	95.06	95.37	95.98	95.89
0.4	80.30	87.31	92.90	93.93	95.57	84.00	88.43	90.34	92.50	90.79	92.47	93.42	94.05	94.65	95.28
0.5	67.38	77.83	85.39	89.00	92.23	71.00	78.06	82.25	86.02	82.35	86.00	87.73	87.80	90.26	91.70

TABLE IV
RECOGNITION RESULTS WITH DIFFERENT FUSION METHODS

Fusion Method	Occlusion Ratio					
	0.0	0.1	0.2	0.3	0.4	0.5
AAv	97.45	97.24	96.73	96.07	95.09	92.04
RB	97.43	97.26	96.40	95.71	94.21	91.03
RWAv	97.52	97.21	96.64	96.15	95.00	92.47
DS	97.49	97.17	96.73	96.18	95.27	92.34

TABLE V
MULTIPLE RECOGNITION RESULTS OF TEN CLASSES TARGET

Occlusion Ratio	0.0	0.1	0.2	0.3	0.4	0.5
Max	97.73	97.48	97.09	96.38	95.95	93.09
Min	97.22	96.84	96.30	95.30	94.30	90.44
Mean	97.52	97.20	96.76	95.91	95.27	91.95
Var	0.011	0.014	0.025	0.050	0.108	0.299

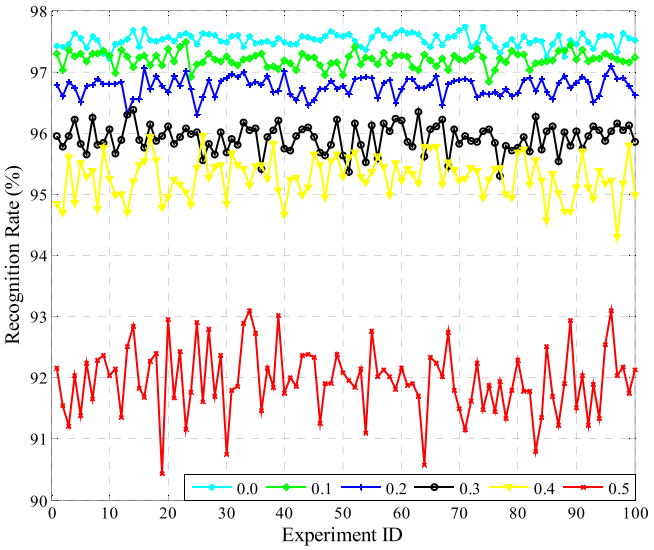


Fig. 6. 100 experimental results of ten classes target.

than 90.00%. This shows that the proposed method not only has good recognition performance, but also has high stability. The reason is that the proposed method overcomes the randomness in single erasure by the fusion strategy.

4) Recognition Performance Comparison: The recognition rates of the proposed method and the comparison algorithms at various occlusion ratios are shown in Table VI. The recognition rate of the proposed method is the average of the above

TABLE VI
RECOGNITION RESULTS OF TEN CLASSES TARGET

Method	Occlusion Ratio					
	0.0	0.1	0.2	0.3	0.4	0.5
LSVM	87.76	76.10	55.68	32.63	24.33	20.25
KSVM	93.94	81.79	57.21	31.88	24.38	20.02
SR	96.80	95.03	89.39	78.06	72.35	58.80
KSR	97.24	92.98	82.67	59.86	49.29	38.08
ISR	96.84	94.81	89.15	77.83	71.41	61.68
WSR	95.97	93.96	92.89	85.36	81.20	72.49
CNN	**99.42**	93.64	80.29	60.41	51.09	38.59
FSVM	92.64	77.01	53.19	29.49	22.98	19.04
FSR	97.16	94.68	86.43	67.08	55.58	42.55
Proposed	97.52	**97.20**	**96.76**	**95.91**	**95.27**	**91.95**

100 experimental results. From these results, many common observations can be raised which are as follows.

1) When the occlusion ratio increases from 0.0 to 0.5, the recognition rates of the proposed method decrease from 97.52% to 91.95%, and the recognition rates of the SR algorithm decrease from 96.80% to 58.80%. The other methods have the same rule. This shows that the larger the occlusion ratio, the worse the recognition performance. The reason is that the larger the occlusion ratio, the more lost information, and the less useful information.
2) When the occlusion ratios are 0.0, 0.1, 0.2, 0.3, 0.4, and 0.5, the recognition rates of the proposed method are 97.52%, 97.20%, 96.76%, 95.91%, 95.27%, and 91.95%, respectively. The highest recognition rates of the comparison algorithms are 99.42% (CNN), 95.03% (SR), 92.89% (WSR), 85.36% (WSR), 81.20% (WSR), and 72.49% (WSR). In addition to the unoccluded case, the proposed method has the highest recognition rate. This means that the recognition performance of the proposed method is better than that of the comparison algorithms under various occlusion ratios. The reason is that multiple random erasures can effectively eliminate the influence of occlusion, and the decision-level fusion can compensate for the erroneous erasure in single erasure. In short, the proposed method can fully utilize the useful information of the unoccluded area during eliminating occlusion.
3) When the occlusion ratio is 0.1, the recognition rate of the proposed method is 21.10%, 15.41%, 2.17%, 4.22%, 2.39%, 3.24%, 3.56%, 20.19%, and 2.52% higher than that of LSVM, KSVM, SR, KSR, ISR, WSR, CNN,

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FSVM, and FSR, respectively. When the occlusion ratio increases to 0.5, the recognition rate of the proposed method is 71.70%, 71.93%, 33.15%, 53.87%, 30.27%, 19.46%, 53.36%, 72.91%, and 49.40% higher than that of the comparison methods. This indicates that the greater the occlusion ratio, the more obvious the advantage of the proposed method.

4) When the occlusion ratio increases from 0.0 to 0.5, the recognition rates of the proposed method decrease by 5.57%. The recognition rates of LSVM, KSVM, SR, KSR, ISR, WSR, CNN, FSVM, and FSR decrease by 67.51%, 73.92%, 38.00%, 59.16%, 35.16%, 23.48%, 60.83%, 73.60%, and 54.61%, respectively. This indicates that the recognition rate of the proposed method decreases the least when the occlusion ratio increases, that is, the proposed method has the best robustness in the occlusion case.
5) In the case of occlusion, the recognition performance of SVM-related methods (LSVM, KSVM, and FSVM) deteriorates dramatically. This means that the SVM method is not suitable for recognizing the occluded target. In SR-related methods, the recognition rate of the proposed method, SR, WSR, and ISR decreases less, while that of the KSR and FSR decreases much more. The reason is that the occlusion area does not diffuse in the first four methods. However, the kernel processing strategy in the KSR method causes the occlusion effect to spread to the entire sample. This diffusion leads to a decrease in recognition performance. Therefore, it is very important to keep the nondiffusion of the occluded area in identifying the occluded target.

It can be seen from the above experimental results that the proposed method has good recognition performance in the case of large erasure ratio, and the fusion strategy has less influence on the recognition performance. Therefore, we use the erasure type XV and the AAv algorithm in the subsequent experiments. In addition, we only focus on the statistical property and the performance of the proposed method.

C. Recognition Under Incomplete Samples

The aspect angle of the target in the MSTAR database is 0°–360°. Therefore, the difference of the aspect angle between adjacent images is 1°–3°. In this case the training samples are sufficient and the dictionary is complete. This means that no matter what the aspect angle of the test sample is, there are always training samples in the dictionary which have little difference from the aspect angle of the test sample. In practice, there may not be enough SAR images. The difference of aspect angles between adjacent SAR images is large. This may lead to a large difference of the aspect angle between the training sample and the test sample. Since the SAR image is sensitive to the aspect angle, this difference can seriously affect the recognition performance. This experiment verifies the recognition performance of the proposed method under the condition that the training samples are insufficient. The experiment also uses the ten classes of the targets as the experimental objects. The images captured at depression angle 17° are used as the training samples. The images captured at depression angle 15° are used as the test samples. We use interval sampling with sampling interval 3 in the training samples to simulate the incomplete dictionary. Suppose the original training samples are $\{\mathbf{y}^1, \mathbf{y}^2, \mathbf{y}^3, \mathbf{y}^4, \ldots\}$, then the training samples after interval sampling are $\{\mathbf{y}^1, \mathbf{y}^4, \ldots\}$. Compared with Table II, the test samples remain unchanged, and the training samples are reduced to one-third. The statistical information of the experimental data is shown in Table VII. The results of 100 experiments are shown in Table VIII and Fig. 7. The recognition rates of various methods are shown in Table IX.

TABLE VII
STATISTICAL INFORMATION OF TEN CLASSES TARGET UNDER INCOMPLETE SAMPLES

Class	Train(17°)	Test(15°)
BMP2	78	195
BTR70	78	196
T72	78	196
BTR60	86	195
2S1	100	274
BRDM2	100	274
D7	100	274
T62	100	273
ZIL131	100	274
ZSU23/4	100	274

TABLE VIII
MULTIPLE RECOGNITION RESULTS OF TEN CLASSES TARGET UNDER INCOMPLETE SAMPLES

Occlusion Ratio	0.0	0.1	0.2	0.3	0.4	0.5
Max	94.11	93.97	92.97	90.92	89.66	85.34
Min	93.14	92.70	91.90	89.49	88.10	82.40
Mean	93.63	93.36	92.39	90.13	89.04	84.12
Var	0.043	0.056	0.048	0.088	0.141	0.344

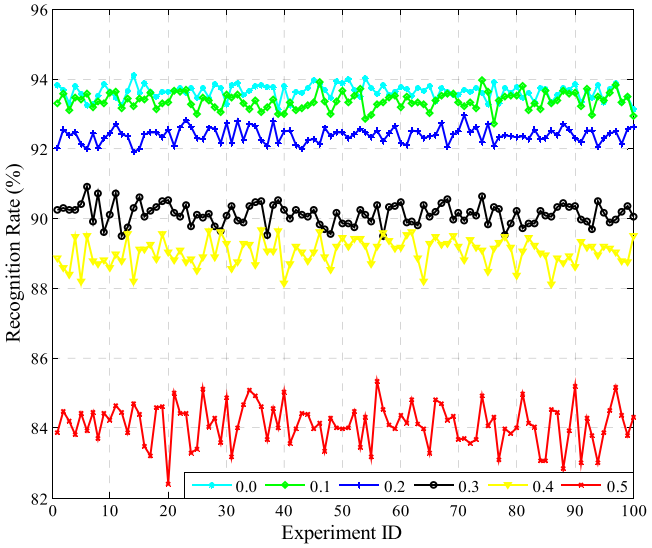


Fig. 7. 100 experimental results of ten classes target under incomplete samples.

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TABLE IX
RECOGNITION RESULTS OF TEN CLASSES TARGET UNDER INCOMPLETE SAMPLES

Method	Occlusion Ratio					
	0.0	0.1	0.2	0.3	0.4	0.5
LSVM	72.30	64.47	49.73	31.54	24.94	21.21
KSVM	81.09	71.25	52.81	31.03	24.58	20.51
SR	93.60	90.29	82.93	68.68	61.11	50.38
KSR	93.46	88.27	77.74	57.86	47.62	34.58
ISR	93.52	90.35	83.20	69.88	63.03	52.18
WSR	92.57	89.15	85.54	74.55	71.11	60.24
CNN	**95.30**	82.85	63.17	43.38	35.88	28.74
FSVM	81.50	66.29	47.41	27.45	21.30	18.05
FSR	93.35	91.40	81.46	62.46	51.48	38.99
Proposed	93.63	**93.36**	**92.39**	**90.13**	**89.04**	**84.12**

It can be seen from Table VIII and Fig. 7 that although the variance of the proposed method has an increasing trend as the occlusion ratio increases, the maximum variance is only 0.344. This indicates that the proposed method has good stability under the condition of incomplete training samples.

When the occlusion ratio increases from 0.0 to 0.5, the recognition rates of the proposed method are 97.52%, 97.20%, 96.76%, 95.91%, 95.27%, and 91.95% in the case of complete dictionary. However, the corresponding recognition rates of the proposed method are 93.63%, 93.36%, 92.39%, 90.13%, 89.04%, and 84.12% in the case of incomplete dictionary. The other methods have the same rule. This shows that the completeness of the dictionary can directly affect the recognition performance. When the occlusion ratio is 0.1, the recognition rate of the proposed method is 93.36%, which is slightly higher than that of FSR (91.40%). When the occlusion ratio increases to 0.5, the recognition rate of the proposed method is 84.12%, which is obviously higher than the comparison method with the best recognition performance (WSR 60.24%). The recognition performance of the proposed method is better than that of the comparison methods in the case of occlusion. The larger the occlusion ratio, the more obvious the advantage of the proposed method. This shows that the proposed method has the best recognition performance and robustness in the case of incomplete dictionary.

D. Recognition Under Random Noise Corruption

The SAR system is inevitably disturbed by various external and internal noises. These noises can destroy the similarity among the images, which seriously affects the performance of the recognition algorithm. This experiment verifies the recognition performance of the proposed method in the case that the test samples are contaminated by noise. The experimental data is the same as Table II. We use the simulation method to generate the noise corrupted images. We corrupt 30% (noise corruption ratio) of randomly chosen pixels from the image by replacing them with independently and identically distributed samples generated from a uniform distribution. The noise corruption image is shown in Fig. 8. The test sample is obtained by using the background image to randomly occlude the noise corrupted image. The results of 100 experiments are shown in Table VIII and Fig. 9. The recognition rates of various methods are shown in Table XI.

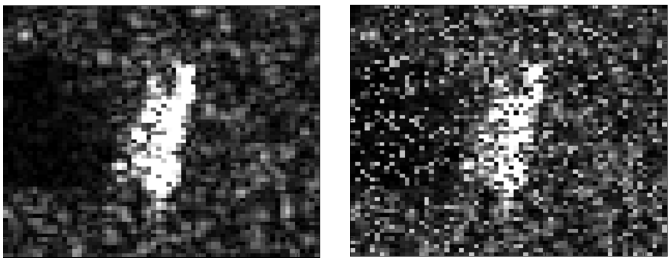


Fig. 8. Example of target with random noise corruption. (Left) Original image. (Right) Image with random noise corruption.

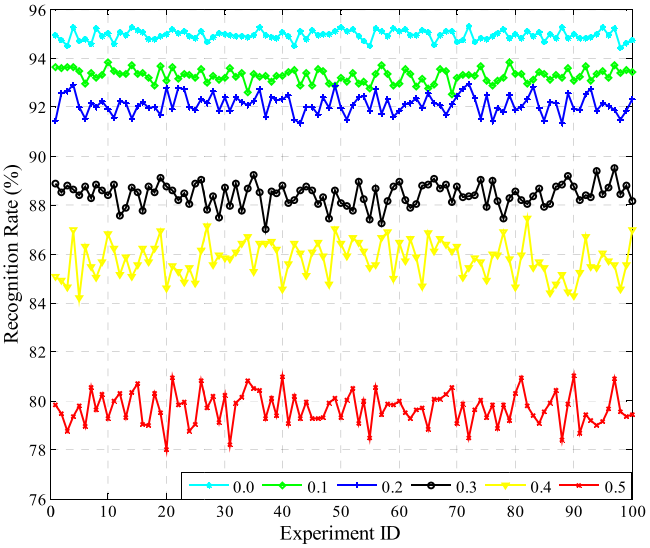


Fig. 9. 100 experimental results of ten classes target under random noise corruption.

TABLE X
MULTIPLE RECOGNITION RESULTS OF TEN CLASSES TARGET UNDER RANDOM NOISE CORRUPTION

Occlusion Ratio	0.0	0.1	0.2	0.3	0.4	0.5
Max	95.31	93.85	92.96	89.51	87.43	81.01
Min	94.42	92.53	91.33	87.03	84.18	77.99
Mean	94.95	93.28	92.13	88.43	85.77	79.69
Var	0.040	0.073	0.157	0.228	0.552	0.424

It can be seen from Table X and Fig. 9 that the smaller the occlusion ratio, the smaller the variance, and the larger the occlusion ratio, the larger the variance. The maximum variance is 0.552, and the absolute difference of the recognition rates at this time is 3.25%. This indicates that the proposed method has good statistical property under the condition of random noise corruption. It can be seen from Tables V and X that the noise corruption increases the variance of the proposed method. That is, the noise corruption can deteriorate the stability of the proposed method.

In the absence and presence of noise, the recognition rates of the proposed method are 97.20% and 93.28% under the condition of occlusion ratio 0.1, and the recognition rates of the proposed method are 95.27% and 85.77% under the condition of occlusion ratio 0.4. This means that the noise corruption can deteriorate the recognition performance of the

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TABLE XI

RECOGNITION RESULTS OF TEN CLASSES TARGET UNDER RANDOM NOISE CORRUPTION

Method	Occlusion ratio					
	0.0	0.1	0.2	0.3	0.4	0.5
LSVM	66.28	51.34	36.14	23.79	20.67	18.70
KSVM	71.28	53.80	37.06	23.62	20.42	18.85
SR	91.48	83.14	71.41	56.02	47.64	40.63
KSR	89.70	75.53	58.44	39.84	32.19	25.07
ISR	92.47	86.41	74.70	56.16	49.37	39.98
WSR	89.42	83.10	72.97	61.25	54.88	46.04
CNN	34.72	32.21	28.13	22.19	20.04	16.82
FSVM	65.08	49.03	34.66	23.23	20.24	18.80
FSR	92.89	81.68	65.54	46.47	37.93	31.99
Proposed	**94.95**	**93.28**	**92.13**	**88.43**	**85.77**	**79.69**

TABLE XII

RECOGNITION RATES OF VARIOUS METHODS WITH NOISE CORRUPTION RATIO 10%

Method	Occlusion Ratio					
	0.0	0.1	0.2	0.3	0.4	0.5
LSVM	87.08	69.73	47.96	28.42	23.62	20.15
KSVM	92.20	75.11	49.84	28.29	22.81	20.13
SR	96.14	92.71	84.85	72.02	64.07	52.20
KSR	96.64	89.21	72.45	49.61	41.47	31.01
ISR	96.23	93.90	86.72	72.85	63.96	54.53
WSR	95.58	92.20	86.10	76.40	69.66	60.55
CNN	33.61	29.20	25.77	23.96	22.47	21.53
FSVM	89.83	69.93	45.89	26.71	21.70	19.12
FSR	96.97	92.85	80.65	59.59	50.07	38.47
Proposed	**97.18**	**96.74**	**96.34**	**95.26**	**93.13**	**90.35**

TABLE XIII

RECOGNITION RATES OF VARIOUS METHODS WITH NOISE CORRUPTION RATIO 50%

Method	Occlusion Ratio					
	0.0	0.1	0.2	0.3	0.4	0.5
LSVM	42.14	32.92	25.87	17.97	17.38	16.28
KSVM	44.02	32.98	24.64	17.76	16.90	15.72
SR	76.55	65.80	55.18	41.12	36.45	29.65
KSR	69.05	53.67	41.58	29.39	26.01	21.85
ISR	80.44	69.60	57.01	42.15	37.66	31.46
WSR	75.62	68.28	59.08	46.70	43.76	35.75
CNN	10.47	10.39	10.27	10.43	10.43	10.27
FSVM	41.42	32.41	24.95	18.42	17.80	17.15
FSR	79.12	61.70	49.17	33.38	30.48	25.04
Proposed	**86.56**	**82.65**	**80.39**	**74.95**	**72.52**	**65.33**

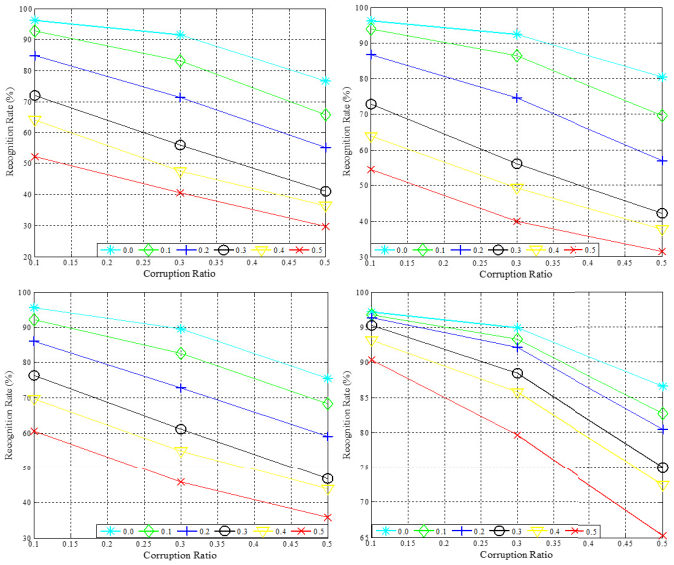


Fig. 10. Recognition rate curves of the SR, ISR, WSR, and the proposed method.

proposed method. The other methods also show the same rule. At various occlusion ratios, the recognition rates of the proposed method are the highest. When the occlusion ratio is 0.0, the recognition rate of the proposed method is 2.06% higher than that of the best comparison method (FSR 92.89%). As the occlusion ratio increases, the difference between the proposed method and the comparison methods increases. The difference increases to 33.65% (the proposed method and WSR) as the occlusion ratio increases to 0.5. This shows that the proposed method has the best recognition performance and robustness in the case of random noise corruption.

In order to more fully evaluate the influence of noise on the proposed method, we added two experiments of noise corruption. The noise corruption ratios in these experiments are 10% and 50%, respectively. The recognition rates of various methods in the case of noise corruption ratio of 10% are shown in Table XII. The recognition rates of various methods in the case of noise corruption ratio of 50% are shown in Table XIII. Under the condition that the noise corruption ratio increases from 10% to 50%, the recognition rate curves of the SR, ISR, WSR, and the proposed method are shown in Fig. 10.

When the noise corruption ratio is 10%, the recognition rates of the proposed method in various occlusion ratios are 97.18%, 96.74%, 96.34%, 95.26%, 93.15%, and 90.35%. At this time, the best recognition rates of the comparison methods are 96.97% (FSR), 93.90% (ISR), 86.72% (ISR), 76.40% (WSR), 69.66% (WSR), and 60.55% (WSR). The recognition rate of the proposed method is significantly higher than that of the comparison methods. When the noise corruption ratio is 50%, the recognition rates of the proposed method in various occlusion ratios are 86.56%, 82.65%, 80.39%, 74.95%, 72.52%, and 65.33%. At this time, the best recognition rates of the comparison methods are 80.44% (ISR), 69.60% (ISR), 59.08% (WSR), 46.70% (WSR), 43.76% (WSR), and 35.75% (WSR). The recognition rates of the proposed method is also significantly better than that of the comparison methods. This shows that the proposed method has the best recognition performance at various levels of noise corruption. When the noise corruption ratio increases from 10% to 50%, the recognition rates of SR, ISR, WSR, and the proposed method gradually decrease. The other methods also show the same rule. This indicates that the level of noise corruption can directly affect the recognition performance of the methods. The more severe the noise corruption, the worse the performance of the recognition method. Therefore, reducing the noise level is an effective way to improve the recognition performance of the method.

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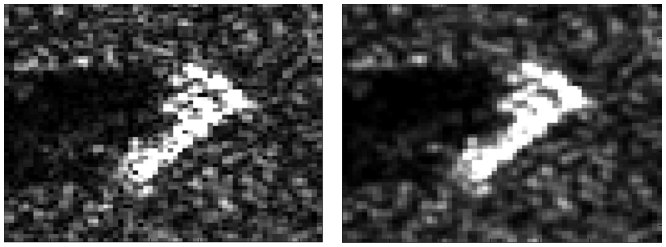


Fig. 11. SAR images with resolutions of 0.3 m × 0.3 m and 0.6 m × 0.6 m.

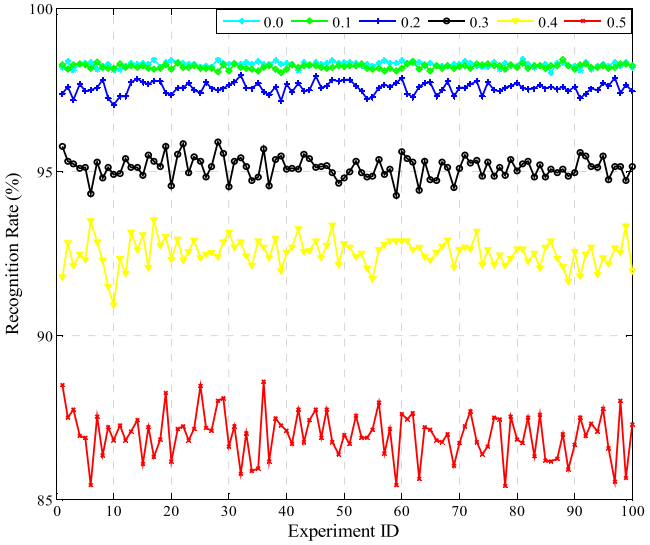


Fig. 12. Recognition rates with resolution of 0.6 m × 0.6 m.

E. Recognition Under Low Resolution Image

The resolution directly determines the detail information of the target contained in the image. The higher the resolution, the more detail information the image contains. The lower the resolution, the less detail information the image contains. Therefore, it is necessary to verify the recognition performance of the proposed method at different resolutions. In this experiment we use the simulation method to obtain SAR images with other resolution. Since the resolution of the SAR image is determined by the bandwidth and the aperture length, we use the method of segmenting the frequency domain to obtain images with different resolutions [38], [39]. First, this method transforms the SAR image into the frequency domain. Then, the data in frequency domain is segmented according to the ratio of the required resolution to the original resolution. Finally, the segmented data is transformed into the image domain to obtain the image with the required resolution. The resolution of the SAR image in the MSTAR data set is 0.3 m × 0.3 m. We obtain the image with the resolution of 0.6 m × 0.6 m through the above simulation method. The images with two resolutions are shown in Fig. 11. We use ten classes target as the experimental objects. The statistical information of this experiment is the same as the SOC experiment. The recognition results of the proposed method repeated 100 experiments are shown in Fig. 12. The statistical information of the recognition rate is shown in Table XIV.

TABLE XIV

STATISTICAL INFORMATION OF RECOGNITION RATES WITH RESOLUTION OF 0.6 m × 0.6 m

Occlusion Ratio	0.0	0.1	0.2	0.3	0.4	0.5
Max	98.45	98.42	97.95	95.91	93.54	88.61
Min	98.03	98.02	97.03	94.28	90.92	85.40
Mean	98.29	98.21	97.57	95.13	92.53	86.97
Var	0.006	0.005	0.032	0.106	0.185	0.484

TABLE XV

RECOGNITION RATES OF VARIOUS METHODS WITH RESOLUTION OF 0.6 m × 0.6 m

Method	Occlusion Ratio					
	0.0	0.1	0.2	0.3	0.4	0.5
LSVM	90.64	72.38	47.32	27.01	22.66	20.13
KSVM	96.03	78.01	51.27	28.49	22.83	20.55
SR	98.14	95.95	86.63	73.48	65.36	53.30
KSR	97.62	89.58	69.68	46.76	37.28	27.47
ISR	98.14	96.60	89.14	75.64	66.41	53.41
WSR	97.79	96.63	92.08	82.88	75.71	64.76
CNN	**99.42**	90.93	75.75	53.20	42.52	32.41
FSVM	95.06	73.61	47.38	26.85	21.71	19.89
FSR	98.19	94.63	77.88	56.29	47.05	35.73
Proposed	98.29	**98.21**	**97.57**	**95.13**	**92.53**	**86.97**

The recognition rates of various methods are shown in Table XV.

When the occlusion ratio increases from 0.0 to 0.5, the variances of the recognition rate are 0.006, 0.005, 0.032, 0.106, 0.185, and 0.484, respectively. This shows that as the occlusion ratio increases, the variance has an increasing trend. The maximum variance of the recognition rate is 0.484. At this time, the difference between the maximum recognition rate and the minimum recognition rate is 3.21%. The recognition rate of the proposed method is relatively stable at various occlusion ratios. This means that the proposed method has good statistical characteristics in the case of low resolution.

As the occlusion ratio increases from 0.0 to 0.5, the highest recognition rates of the comparison methods are 99.42% (CNN), 96.63% (WSR), 92.08% (WSR), 82.88% (WSR), 75.71% (WSR), and 64.76% (WSR). Except for the unoccluded case, the WSR has the highest recognition rate among all comparison methods. In the case of the unocclusion, the recognition rate of the WSR is 97.79%, which is only slightly lower than the highest recognition rate (99.42%) of the comparison methods. This means that the WSR has the best recognition performance in the case of low resolution image. Therefore, we only compare the recognition performance of the WSR and the proposed method. When the occlusion ratio increases from 0.0 to 0.5, the recognition rates of the proposed method are 98.29%, 98.21%, 97.57%, 95.13%, 92.53%, and 86.97%. The recognition rate of the proposed method is higher than that of the WSR. The difference of the recognition rate between the two methods is 0.50% in the case of occlusion ratio 0.0. However, the difference increases to 22.21% in the case of occlusion ratio 0.5. This shows that the proposed method has the best recognition performance. Moreover, the larger the occlusion ratio, the more obvious the advantage of the proposed method.

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TABLE XVI

STATISTICAL INFORMATION OF FOUR CLASSES TARGET UNDER CONFIGURATION VARIANCE

	Depr	BMP2	T72	BTR60	T62
Train	17°	233(sn-9563)	232(sn-132)	256	299
Test	15°	196(sn-9566) 196(sn-c21)	195(sn-812) 191(sn-s7)	195	273

TABLE XVII

MULTIPLE RECOGNITION RESULTS OF FOUR CLASSES TARGET UNDER CONFIGURATION VARIANCE

Occlusion Ratio	0.0	0.1	0.2	0.3	0.4	0.5
Max	89.72	86.78	84.55	80.69	80.55	77.10
Min	88.14	85.43	82.50	78.37	76.89	73.17
Mean	88.99	85.96	83.41	79.77	78.90	74.89
Var	0.070	0.062	0.154	0.192	0.549	0.559

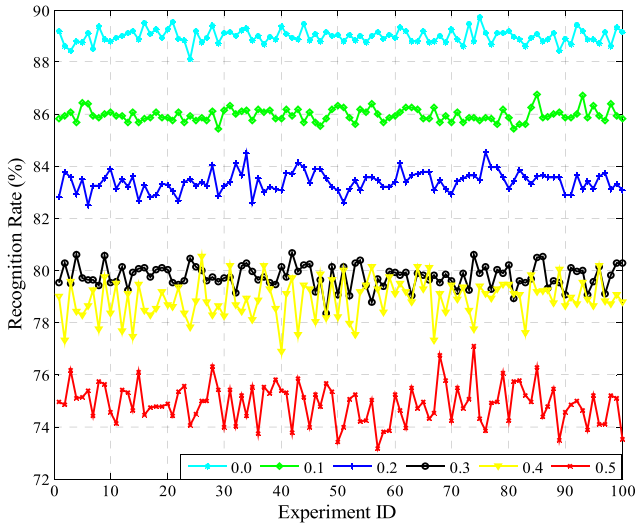


Fig. 13. 100 experimental results of four classes target under configuration variance.

F. Recognition Under Configuration Variance

With the development of the technology and the change of the application scenarios, the target may have multiple serial numbers. The target with different serial numbers may have different appearance. This can lead to the SAR image changes. This experiment evaluates the recognition performance of the proposed method under the condition of configuration variance. This experiment use four classes of targets (BMP2, BTR60, T72, and T62) as the experimental objects. The BTR60 and T62 have only one serial number. The BMP2 and T72 have multiple serial numbers. The serial numbers sn-c9563 (BMP2) and sn-132 (T72) are used as the training samples. The serial numbers sn-c9566, sn-c21 (BMP2) and sn-812, sn-s7 (T72) are used as the test samples. The images captured at depression angle 17° are used as the training samples. The images captured at depression angle 15° are used as the test samples. The statistical information of the experimental data is shown in Table XVI. The results of 100 experiments are shown in Table XVII and Fig. 13. The recognition rates of various methods are shown in Table XVIII.

TABLE XVIII

RECOGNITION RESULTS OF FOUR CLASSES TARGET UNDER CONFIGURATION VARIANCE

Method	Occlusion ratio					
	0.0	0.1	0.2	0.3	0.4	0.5
LSVM	88.60	77.75	62.68	41.48	36.34	30.73
KSVM	90.42	82.12	62.69	41.43	35.91	32.00
SR	90.83	84.68	78.04	64.80	62.75	51.73
KSR	86.20	81.32	79.00	67.31	62.83	51.76
ISR	87.70	84.71	75.90	66.26	62.85	57.33
WSR	89.82	84.69	81.32	71.71	67.14	59.64
CNN	**95.83**	85.00	70.38	48.56	43.17	31.22
FSVM	92.95	78.21	61.05	41.10	35.70	31.87
FSR	91.09	84.76	77.81	63.24	60.07	47.05
Proposed	88.99	**85.96**	**83.41**	**79.77**	**78.90**	**74.89**

It can be seen from Table XVII and Fig. 13 that the variance of the proposed method has an increasing trend as the occlusion ratio increases. Although the maximum variance is 0.559, the absolute difference of the recognition rates at this time is 3.93%. Even with the worst recognition performance, the recognition rate at the occlusion ratio of 0.5 is 73.17%, which is also much larger than the best recognition result of the comparison methods under the corresponding occlusion ratio (WSR 59.64%). This means that the proposed method has good statistical property in the case of configuration variance.

When the occlusion ratio increases from 0.0 to 0.5, the recognition rates of the proposed method are 88.99%, 85.96%, 83.41%, 79.77%, 78.90%, and 74.89%, respectively. The recognition rates of the highest competitor are 95.83% (CNN), 85.00% (CNN), 81.32% (WSR), 71.71% (WSR), 67.14% (WSR), and 59.64% (WSR) in the corresponding occlusion ratio. Although the recognition rate of the proposed method is lower than that of the CNN, FSVM, KSVM, SR, WSR, and FSR in the case of no occlusion, the proposed method has the highest recognition rate in various occlusion ratios. The difference of recognition rates between the proposed method and the nearest competitor reaches 15.25% at the occlusion ratio of 0.5. This means that the greater the occlusion ratio, the more obvious the advantage of the proposed method. The experimental results indicate that the proposed method has the best robustness in the case of configuration variance.

G. Recognition Under Depression Angle Variance

In the previous experiments, the depression angle of the training sample is 17°, while the depression angle of the test sample is 15°. This means that the difference of depression angle between the training sample and the test sample is small. However, the depression angle may vary widely in practice. Since the SAR image is sensitive to the depression angle, the change of depression angle can lead to the change of image. This experiment verifies the recognition performance of the proposed method under the condition of depression angle variance. The experiment uses three classes of targets (2S1, BRDM2, ZSU23/4) as the experimental objects. The images captured at depression angle 17° are used as the training samples. The images captured at depression angle

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TABLE XIX

STATISTICAL INFORMATION OF THREE CLASSES TARGET UNDER DEPRESSION ANGLE VARIANCE

	Depr	2S1	BRDM2	ZSU23/4
Train	17°	299	298	299
Test	30°	288	287	288

TABLE XX

MULTIPLE RECOGNITION RESULTS OF THREE CLASSES TARGET UNDER DEPRESSION ANGLE VARIANCE

Occlusion Ratio	0.0	0.1	0.2	0.3	0.4	0.5
Max	97.92	90.51	84.61	75.35	69.79	62.85
Min	96.53	88.77	81.37	71.64	64.24	58.91
Mean	97.22	89.81	82.97	73.40	67.06	60.71
Var	0.059	0.120	0.593	0.844	1.591	0.519

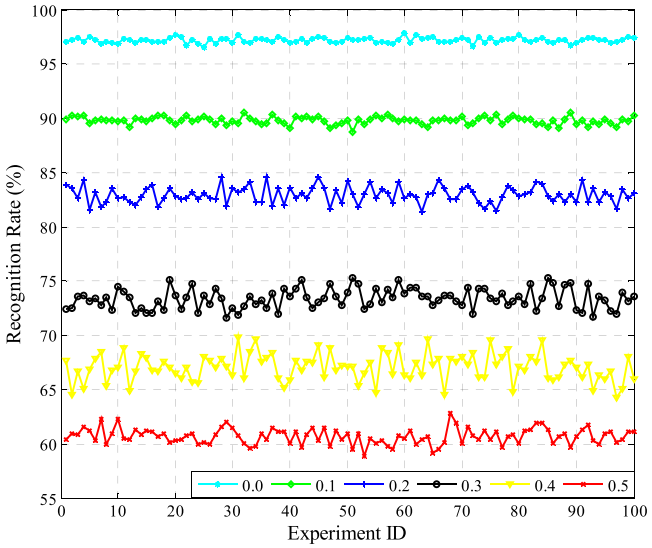


Fig. 14. 100 experimental results of three classes target under depression angle variance.

TABLE XXI

RECOGNITION RESULTS OF THREE CLASSES TARGET UNDER DEPRESSION ANGLE VARIANCE

Method	Occlusion ratio					
	0.0	0.1	0.2	0.3	0.4	0.5
LSVM	89.00	66.78	51.39	38.66	36.57	35.07
KSVM	92.59	69.79	52.78	39.00	36.23	35.07
SR	98.26	86.46	69.10	53.01	48.73	43.75
KSR	98.96	86.57	69.91	49.65	43.29	38.77
ISR	97.33	89.35	73.38	56.36	50.34	46.29
WSR	96.99	87.04	77.20	60.18	54.51	50.93
CNN	97.45	84.24	70.10	58.63	54.23	51.10
FSVM	85.88	61.57	46.88	37.04	35.65	34.49
FSR	**99.88**	88.43	69.56	51.04	44.79	39.47
Proposed	97.22	**89.81**	**82.97**	**73.40**	**67.06**	**60.71**

30° are used as the test samples. The statistical information of the experimental data is shown in Table XIX. The results of 100 experiments are shown in Table XX and Fig. 14. The recognition rates of various methods are shown in Table XXI.

It can be seen from Table XX and Fig. 14 that the variance is small under small occlusion ratio, and the variance is large under large occlusion ratio. This indicates that the stability of the proposed method is high under the condition of small occlusion ratio. Although the absolute difference is 5.55% at the maximum value of the variance, the minimum recognition rate of the proposed method (64.24%) is still much larger than the highest recognition rate of the comparison method (WSR 54.51%) under the corresponding occlusion ratio.

When the occlusion ratio increases from 0.0 to 0.5, the highest recognition rates of the comparison methods are 99.88% (FSR), 89.35% (ISR), 77.20% (WSR), 60.18% (WSR), 54.51% (WSR), and 51.10% (CNN), respectively. Except for the condition of small occlusion ratio, the WSR is superior to the other comparison methods. This means that the WSR method has better performance and robustness under the condition of depression angle variance. The recognition rates of the proposed method under various occlusion ratios are 97.22%, 89.81%, 82.97%, 73.40%, 67.06%, and 60.71%, respectively. The difference between the proposed method and the WSR is 0.23%, 2.77%, 5.77%, 13.22%, 12.55%, and 9.78%, respectively. The proposed method has better recognition performance than that of the WSR. This proves that the proposed method has the best performance and robustness in the case of depression angle variance.

V. CONCLUSION

In this article, the recognition method using fusion strategy of SR based on randomly erased images is proposed to recognize the partially occluded SAR target. The proposed method eliminates the effect of the occlusion by random erasure. The fusion strategy is used to eliminate the under-erasure of the occluded area and the erroneous erasure of the unoccluded area. Although the erased area in single erasure is random, the proposed method uses the fusion strategy to make the recognition performance stable. The experimental results show that the proposed method has good recognition performance at large erasure ratio. In the case of the occlusion, the recognition performance of the proposed method is significantly better than that of the comparison algorithms. The more the occlusion, the more obvious the advantage. In the case of no occlusion, the recognition performance of the proposed method is similar to that of most comparison algorithms, even slightly higher than that of some comparison algorithms. However, the recognition performance of the proposed method is slightly lower than that of the CNN under the condition of no occlusion. This will be the author's next research focus.

REFERENCES

[1] S.-W. Chen, X.-S. Wang, and S.-P. Xiao, "Urban damage level mapping based on co-polarization coherence pattern using multitemporal polarimetric SAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 8, pp. 2657–2667, Aug. 2018.

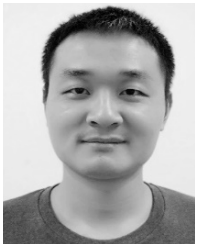
[2] K. El-Darymli, E. W. Gill, P. Mcguire, D. Power, and C. Moloney, "Automatic target recognition in synthetic aperture radar imagery: A State-of-the-art review," *IEEE Access*, vol. 4, pp. 6014–6058, 2016.

[3] S.-W. Chen, X.-S. Wang, and M. Sato, "Urban damage level mapping based on scattering mechanism investigation using fully polarimetric SAR data for the 3.11 East Japan earthquake," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 6919–6929, Dec. 2016.

[4] T. D. Ross, S. W. Worrell, V. J. Velten, J. C. Mossing, and M. L. Bryant, "Standard SAR ATR evaluation experiments using the MSTAR public release data set," in *Proc. Algorithms Synth. Aperture Radar Imag. V*, Sep. 1998, pp. 566–573.

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[5] R. Zhang and M. Zhang, "SAR target recognition based on active contour without edges," *J. Syst. Eng. Electron.*, vol. 28, no. 2, pp. 276–281, Apr. 2017.
[6] S. Papson and R. M. Narayanan, "Classification via the shadow region in SAR imagery," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 48, no. 2, pp. 969–980, Apr. 2012.
[7] Z. He, J. Lu, and G. Kuang, "A fast SAR target recognition approach using PCA features," in *Proc. 4th Int. Conf. Image Graph. (ICIG)*, Sichuan, China, Aug. 2007, pp. 580–585.
[8] A. K. Mishra and T. Motaung, "Application of linear and nonlinear PCA to SAR ATR," in *Proc. 25th Int. Conf. Radioelektronika (RADIOELEKTRONIKA)*, Apr. 2015, pp. 349–354.
[9] Z. Cui, Z. Cao, J. Yang, J. Feng, and H. Ren, "Target recognition in synthetic aperture radar images via non-negative matrix factorisation," *IET Radar, Sonar Navigat.*, vol. 9, no. 9, pp. 1376–1385, Dec. 2015.
[10] A. Karine, A. Toumi, A. Khenchaf, and M. El Hassouni, "Target recognition in ISAR images based on relative phases of complex wavelet coefficients and sparse classification," in *Proc. 4th Int. Conf. Adv. Technol. Signal Image Process. (ATSIP)*, Sousse, Tunisia, Mar. 2018, pp. 1–5.
[11] S. Ma, Q. Gong, and J. Zhang, "Space target recognition based on 2-D wavelet transformation and KPCA," in *Proc. IEEE 3rd Int. Conf. Commun. Softw. Netw.*, Xi'an, China, May 2011, pp. 516–520.
[12] G. Dong, G. Kuang, N. Wang, and W. Wang, "Classification via sparse representation of steerable wavelet frames on Grassmann manifold: Application to target recognition in SAR image," *IEEE Trans. Image Process.*, vol. 26, no. 6, pp. 2892–2904, Jun. 2017.
[13] G. Dong, N. Wang, G. Kuang, and H. Qiu, "Sparsity and low-rank dictionary learning for sparse representation of monogenic signal," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 1, pp. 141–153, Jan. 2018.
[14] G. Jones and B. Bhanu, "Recognition of articulated and occluded objects," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 21, no. 7, pp. 603–613, Jul. 1999.
[15] G. Jones and B. Bhanu, "Recognizing occluded objects in SAR images," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 37, no. 1, pp. 316–328, Jan. 2001.
[16] B. Ding, G. Wen, C. Ma, and X. Yang, "An efficient and robust framework for SAR target recognition by hierarchically fusing global and local features," *IEEE Trans. Image Process.*, vol. 27, no. 12, pp. 5983–5995, Dec. 2018.
[17] B. Ding, G. Wen, X. Huang, C. Ma, and X. Yang, "Target recognition in synthetic aperture radar images via matching of attributed scattering centers," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 7, pp. 3334–3347, Jul. 2017.
[18] J. Chen, B. Zhang, and C. Wang, "Backscattering feature analysis and recognition of civilian aircraft in TerraSAR-X images," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 4, pp. 796–800, Apr. 2015.
[19] C. Clemente, J. J. Soraghan, I. Proudler, A. De Maio, L. Pallotta, and A. Farina, "Pseudo-Zernike-based multi-pass automatic target recognition from multi-channel synthetic aperture radar," *IET Radar, Sonar Navigat.*, vol. 9, no. 4, pp. 457–466, Apr. 2015.
[20] Q. Zhao and J. C. Principe, "Support vector machines for SAR automatic target recognition," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 37, no. 2, pp. 643–654, Apr. 2001.
[21] C. Tao, S. Chen, Y. Li, and S. Xiao, "PolSAR land cover classification based on roll-invariant and selected hidden polarimetric features in the rotation domain," *Remote Sens.*, vol. 9, no. 7, p. 660, 2017.
[22] J. Wright, A. Yang, A. Ganesh, S. Sastry, and Y. Ma, "Robust face recognition via sparse representation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 31, no. 2, pp. 210–227, Feb. 2009.
[23] J. Wang, C. Lu, M. Wang, P. Li, S. Yan, and X. Hu, "Robust face recognition via adaptive sparse representation," *IEEE Trans. Cybern.*, vol. 44, no. 12, pp. 2368–2378, Dec. 2014.
[24] M. Li, Y. Guo, M. Li, G. Luo, and X. Kong, "Coupled dictionary learning for target recognition in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 6, pp. 791–795, Jun. 2017.
[25] A. Karine, A. Toumi, A. Khenchaf, and M. El Hassouni, "Target recognition in radar images using weighted statistical dictionary-based sparse representation," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2403–2407, Dec. 2017.
[26] Z. Zhou, Z. Cao, and Y. Pi, "Subdictionary-based joint sparse representation for SAR target recognition using multilevel reconstruction," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6877–6887, Sep. 2019.
[27] M. Liu, S. Chen, J. Wu, F. Lu, X. Wang, and M. Xing, "SAR target configuration recognition via two-stage sparse structure representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2220–2232, Apr. 2018.
[28] N. Wang, Y. Wang, H. Liu, Q. Zuo, and J. He, "Feature-fused SAR target discrimination using multiple convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1695–1699, Oct. 2017.
[29] S. Chen, H. Wang, F. Xu, and Y.-Q. Jin, "Target classification using the deep convolutional networks for SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4806–4817, Aug. 2016.
[30] Q. Song and F. Xu, "Zero-shot learning of SAR target feature space with deep generative neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2245–2249, Dec. 2017.
[31] J. Pei, Y. Huang, W. Huo, Y. Zhang, J. Yang, and T.-S. Yeo, "SAR automatic target recognition based on multiview deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2196–2210, Apr. 2018.
[32] B. Ding and G. Wen, "Exploiting multi-view SAR images for robust target recognition," *Remote Sens.*, vol. 9, no. 11, p. 1150, 2017.
[33] Z. He, C. Gao, H. Xiao, and Z. Tian, "Adaptive weighting based on subimage sparse model for SAR occluded target recognition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 8, pp. 2976–2988, Aug. 2019.
[34] H. Zhang, N. M. Nasrabadi, Y. Zhang, and T. S. Huang, "Multi-view automatic target recognition using joint sparse representation," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 48, no. 3, pp. 2481–2497, Jul. 2012.
[35] Z. Tian, L. Wang, R. Zhan, J. Hu, and J. Zhang, "Classification via weighted kernel CNN: Application to SAR target recognition," *Int. J. Remote Sens.*, vol. 39, no. 23, pp. 9249–9268, Dec. 2018.
[36] B. Ding, G. Wen, C. Ma, and X. Yang, "Decision fusion based on physically relevant features for SAR ATR," *IET Radar, Sonar Navigat.*, vol. 11, no. 4, pp. 682–690, Apr. 2017.
[37] R. Huan and Y. Pan, "Decision fusion strategies for SAR image target recognition," *IET Radar, Sonar Navigat.*, vol. 5, no. 7, p. 747, 2011.
[38] B. Ding and G. Wen, "Target recognition of SAR images based on multi-resolution representation," *Remote Sens. Lett.*, vol. 8, no. 11, pp. 1006–1014, Nov. 2017.
[39] B. Ding, G. Wen, J. Zhong, C. Ma, and X. Yang, "A robust similarity measure for attributed scattering center sets with application to SAR ATR," *Neurocomputing*, vol. 219, pp. 130–143, Jan. 2017.



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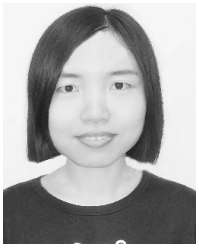
His research interests include machine learning, pattern recognition, and artificial intelligence.



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